RESEARCH PLAN

Aggregating User Preferences while Ensuring Equity, Diversity, and Inclusion using Graph Summarization

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1. CONTEXT AND OBJECTIVES

Preference aggregation in recommender systems, voting platforms, and decision-making processes often fails to preserve equity, diversity, and inclusion (EDI), leading to outcomes that favor majority opinions while marginalizing minority voices. This research addresses this gap by leveraging graph summarization techniques to efficiently aggregate user preferences while explicitly maintaining EDI principles throughout the process.

The research objectives are to:

- Develop a graph-based representation of user preferences capturing individual preferences and their relationships
- Design graph summarization algorithms that preserve EDI metrics (fairness indices, diversity scores, inclusion ratios)
- Implement and evaluate the approach using real-world datasets (MovieLens, product reviews)
- Compare EDI preservation against traditional methods (Borda, Condorcet, ranked choice voting)
- Produce a research paper documenting the methodology and results

2. METHODOLOGY

The research adopts a five-phase approach combining declarative and data-driven methodologies:

Phase 1 - Literature Review (May): Survey existing preference aggregation and graph summarization methods; investigate EDI frameworks and algorithmic fairness metrics; familiarize with tools (Python, NetworkX, Neo4j)

Phase 2 - Model Design (June): Design graph-based preference representation with nodes (users/items) and weighted edges (preferences); define formal EDI metrics for equity, diversity, and inclusion preservation

Phase 3 - Algorithm Development (July): Design summarization algorithms using community detection, graph coarsening, and weighted aggregation; develop constraint-based optimization formulations integrating EDI requirements

Phase 4 - Implementation (August): Implement algorithms in Python; collect benchmark datasets; develop baseline comparison methods

Phase 5 - Evaluation (September): Conduct experiments measuring accuracy, EDI preservation, efficiency, and robustness; compare against baselines; perform statistical analysis

Phase 6 - Documentation (October): Write research paper; prepare final presentation; document code with examples

3. EXPECTED OUTCOMES

The internship will deliver: (1) a novel EDI-aware graph summarization framework with theoretical foundations, (2) open-source Python implementation with documentation, (3) comprehensive experimental evaluation demonstrating EDI preservation compared to traditional methods, and (4) a research paper suitable for submission to AI/data science conferences (AAAI, IJCAI, KDD) or social computing venues (CSCW, ICWSM).

4. TIMELINE

Month	Phase	Key Deliverables
May	Literature Review	Annotated bibliography, technical setup
June	Model Design	Graph model specification, EDI metrics
July	Algorithm Dev.	Algorithm descriptions, complexity analysis
August	Implementation	Functional prototype, baseline methods
September	Evaluation	Experimental results, comparisons
October	Documentation	Research paper, presentation, code

5. RELEVANCE TO MY BACKGROUND

This project aligns excellently with my technical expertise and academic background. My recent work on the **GDB4SmartGrids project** involved extensive use of Neo4j graph databases and Julia programming to optimize smart grid configurations for 10,000 buildings in Versailles, demonstrating potential 7-12% cost savings for collective self-consumption. This experience provided me with strong foundations in graph data modeling, query optimization, and handling large-scale graph structures—skills directly applicable to this research.

My Master 2 DataScale coursework has equipped me with comprehensive knowledge in spatial databases (PostgreSQL/PostGIS), data mining, and algorithmic complexity analysis. Through projects involving mobility data analysis and geographic data processing, I have developed strong skills in formulating optimization problems and implementing efficient algorithms. Additionally, my telecommunications background from École Supérieure Polytechnique and professional experience at Orange Senegal provides practical understanding of large-scale systems and user-centric design. My proficiency in Python, SQL/NoSQL databases, and data processing tools, combined with my achievement of 4th place at the 2024 Huawei ICT Competition, positions me well to implement and evaluate the proposed EDI-aware framework efficiently.

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